

Chapter 4: Nowcasting Mixed-Frequency Bitemporal High Priority Macroeconomic Time Series

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1 Introduction

This document details on asset specific data infrastructure specifically to handle only data engineering to serve directional mid frequency U.S. Equity forecasting. Economic variables exist in a complex system that is hard to determine using a closed function, even running Conditional Independence Tests. Due to the instability of Conditional Independence Tests to produce the similar graphs due to the conditioning set property, Variable Auto Regression for forecasting using weight lag edges derived from Conditional Independence Tests was dropped and substituted for a Nowcasting Kalman Filter.

2 Results

2.1 Run configuration

Run 75aaef9d evaluates 16 macroeconomic series over 7 walk-forward folds spanning June 2020 to June 2026. The mixed-frequency dynamic factor model uses 3 factor(s) with lag order 1 at 1MS resolution, trained on a 72-month context window. Each nowcast is scored against a forward-fill baseline by relative RMSE,

$$\text{rRMSE} = \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{baseline}}}, \quad \text{skill} = 1 - \text{rRMSE},$$

so positive skill indicates the model improves on simply carrying the last published observation forward.

2.2 Aggregate outcome

Across the 16 series, 4 improve on the baseline, 2 tie it and 10 fall short, for a mean skill of -0.1045 and a median of -0.0032. Mean relative RMSE is 1.1045.

2.3 Per-series diagnostics

2.4 Model fit

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Nowcast skill by series and fold

Blue beats the forward-fill baseline, red loses to it, neutral is a tie. Colour clipped at ± 0.36 (11 cells beyond); printed values are exact.

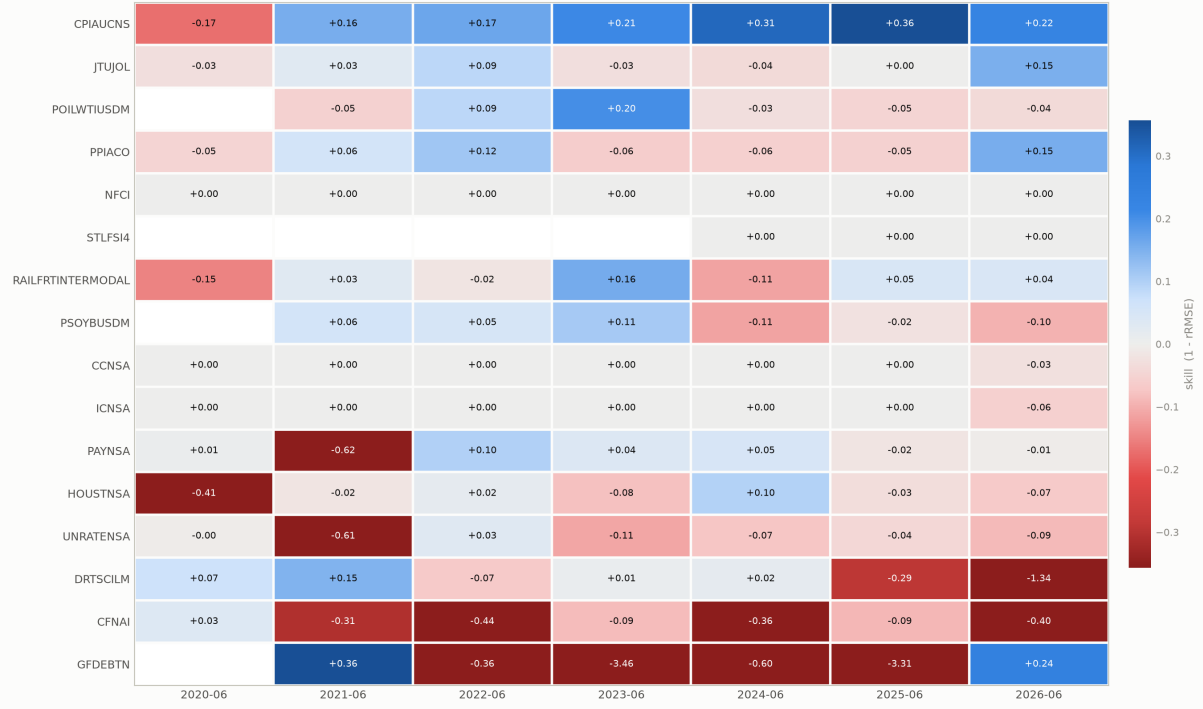


Figure 1: Nowcast skill by series and fold. Blue indicates the model beats the forward-fill baseline, red that it loses; the neutral midpoint is a tie.

Series	Code	Signal share	rRMSE	Skill	Latency (mo)
CPIAUCNS	5	0.419	0.822	+0.178	1.38
JTUJOL	5	0.234	0.976	+0.024	2.24
POILWTIUSDM	5	0.376	0.981	+0.019	2.65
PPIACO	5	0.422	0.985	+0.015	1.38
NFCI	2	0.351	1.000	+0.000	0.17
STLFSI4	2	0.156	1.000	+0.000	0.50
RAILFRTINTERMODAL	5	0.113	1.000	-0.000	2.58
PSOYBUSDM	5	0.094	1.002	-0.002	2.65
CCNSA	5	0.407	1.004	-0.004	0.39
ICNSA	5	0.348	1.008	-0.008	0.16
PAYNSA	5	0.422	1.065	-0.065	1.13
HOUSTNSA	5	0.291	1.071	-0.071	1.58
UNRATENSA	2	0.394	1.128	-0.128	1.13
DRTSCILM	8	0.358	1.208	-0.208	1.13
CFNAI	2	0.832	1.236	-0.236	1.74
GFDEBTN	5	0.317	2.186	-1.186	5.12

Table 1: Per-series diagnostics averaged over folds. “Code” is the FRED-MD style stationarity transformation applied; “signal share” is the common-factor R^2 ; “latency” is the mean publication lag in months.

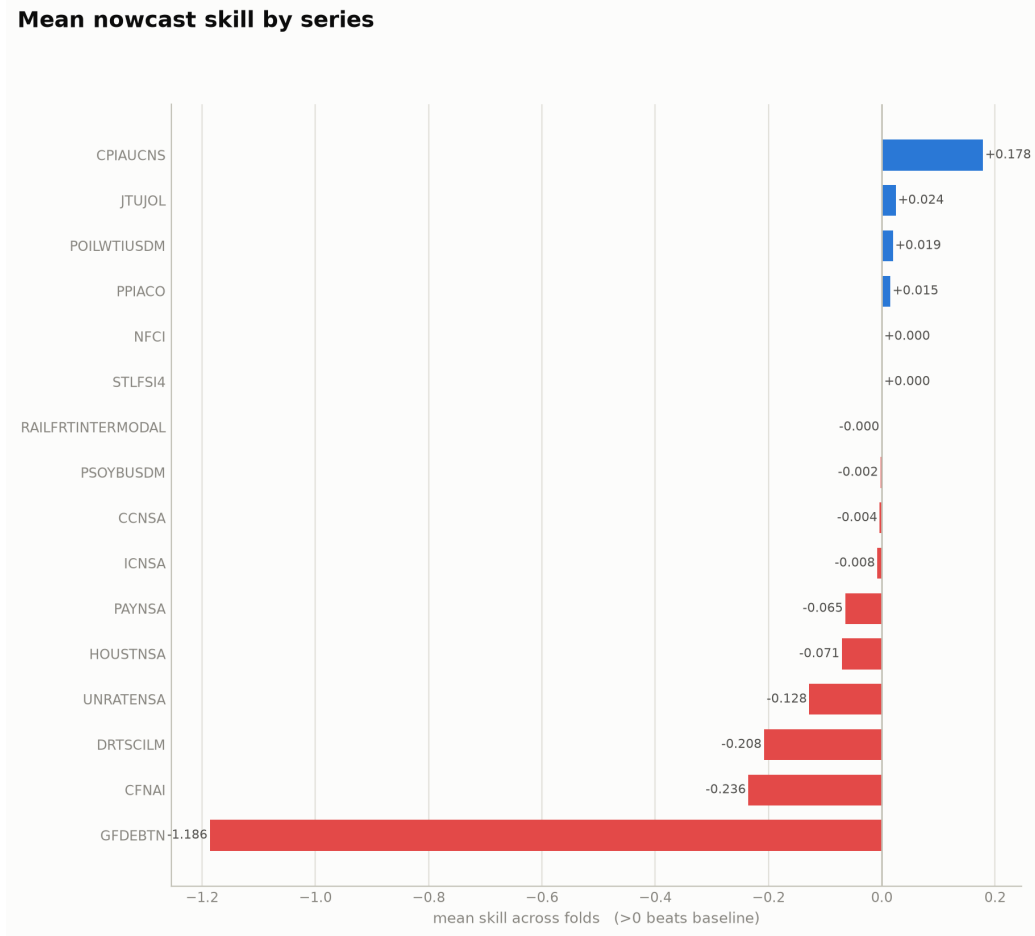


Figure 2: Mean skill by series across all folds.

Fold	N	EM iter.	Log-lik.	AIC	BIC	Factor ρ
2020-06	53	681	-651.8	1,435.7	1,565.7	+0.638, +0.144, -0.133
2021-06	60	736	-737.5	1,637.0	1,806.7	+0.679, +0.243, +0.112
2022-06	60	319	-775.8	1,713.7	1,883.3	+0.276, +0.065, -0.073
2023-06	60	392	-770.6	1,703.3	1,872.9	+0.286, +0.103, +0.064
2024-06	60	203	-812.9	1,797.8	1,977.9	+0.323, -0.117, +0.197
2025-06	60	279	-822.8	1,817.5	1,997.7	+0.322, -0.130, +0.294
2026-06	60	784	-973.7	2,119.4	2,299.5	+0.658, +0.335, +0.513

Table 2: State-space fit statistics per fold.

Nowcast against realised level

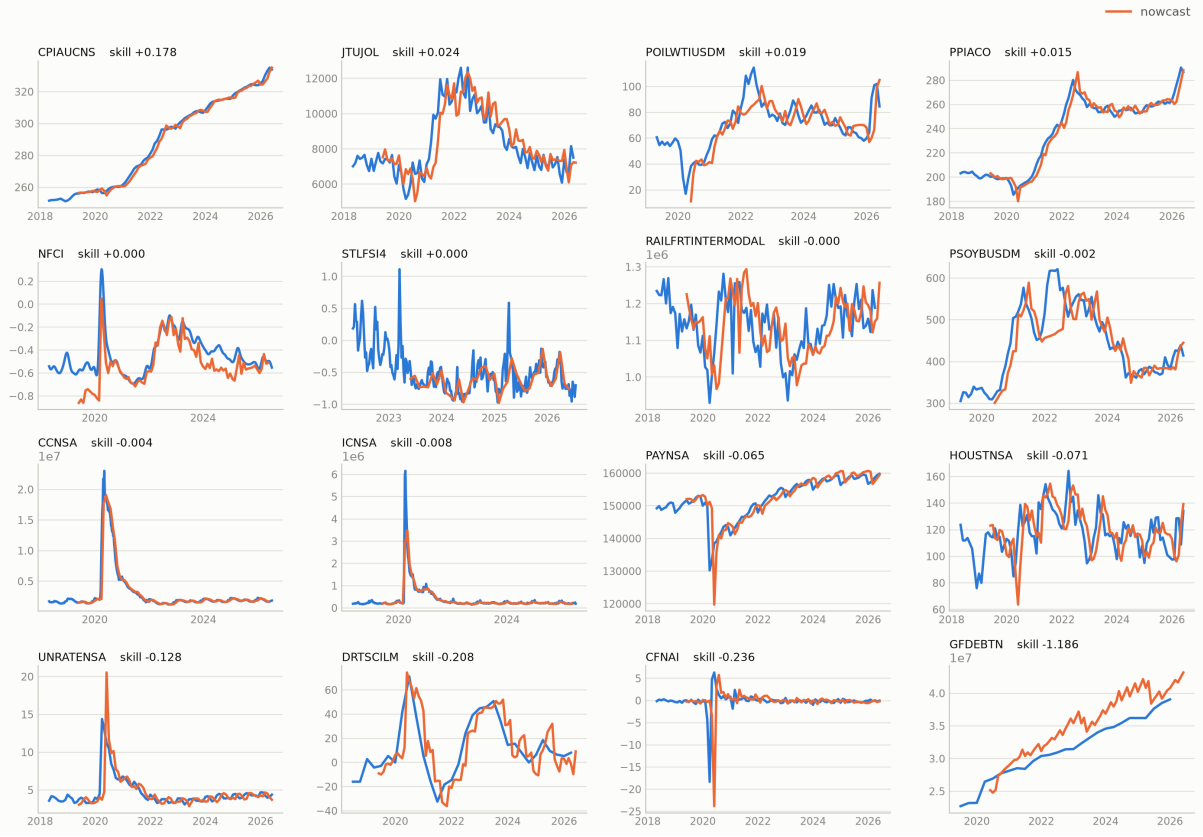


Figure 3: Nowcast against the realised level for each series.

Model fit across folds

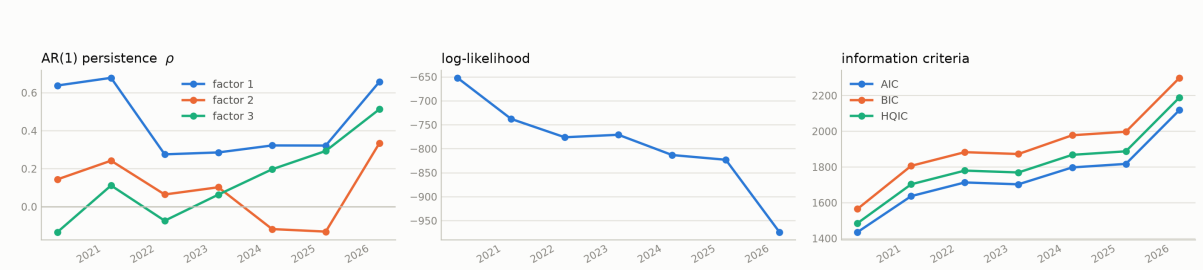


Figure 4: Factor persistence and fit statistics across folds.

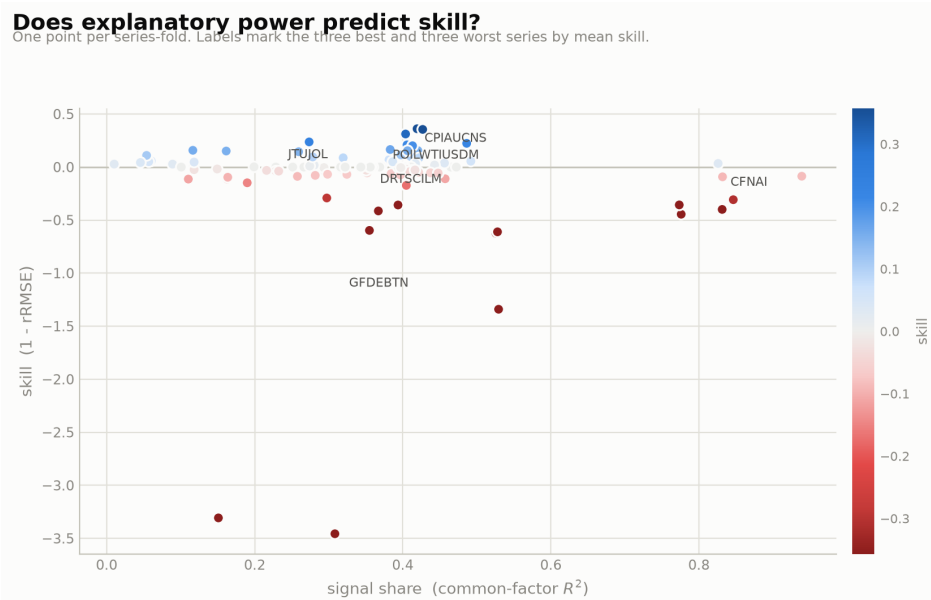


Figure 5: Common-factor explanatory power against realised skill, one point per series-fold.